Pneumonia Classification from Pediatric Chest X-Ray Images: Baseline CNN, Frozen and Fine-Tuned MobileNetV2

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***Abstract*—This study compares three deep-learning configurations for binary classification of pediatric chest X-ray images as Normal or Pneumonia: a baseline convolutional neural network (CNN) trained from scratch, MobileNetV2 with a frozen ImageNet-pretrained feature extractor, and a fine-tuned MobileNetV2 model. Experiments use PneumoniaMNIST from MedMNIST at a configured input size of 224×224 pixels. The official splits contain 4,708 training, 524 validation, and 624 test images. At a classification threshold of 0.5, the fine-tuned MobileNetV2 achieved test accuracy of 0.885, F1-score of 0.915, and ROC-AUC of 0.983, compared with 0.848, 0.891, and 0.970 for the frozen model, and 0.803, 0.857, and 0.893 for the baseline CNN. Fine-tuning reduced false positives among Normal images from 92 to 68 relative to the frozen model while maintaining high pneumonia recall. Bootstrap confidence intervals and paired comparisons were also examined. A validation-selected threshold improved test specificity for the fine-tuned model from 0.709 to 0.722 with a small increase in accuracy. Calibration analysis showed mixed effects and is treated cautiously because the exported results do not document which split was used to fit the temperature parameter. This is an educational benchmark study, not a clinically validated diagnostic system.**

***Index Terms*—pediatric chest X-ray, pneumonia classification, convolutional neural network, transfer learning, MobileNetV2, fine-tuning, threshold selection, calibration**

I. Introduction

Pneumonia is a respiratory infection that can cause inflammation in the lungs and remains an important health concern among children [1]. Chest radiography is frequently used as part of the assessment of suspected pneumonia. Interpreting radiographs requires clinical expertise, and computer-aided image analysis has therefore been investigated as a way to support research into automated classification.

Deep learning, particularly convolutional neural networks (CNNs), has been widely studied for medical image analysis. Kermany *et al.* demonstrated the potential of image-based deep learning for identifying pediatric pneumonia from chest X-ray images [2]. Model performance can depend on the dataset, preprocessing, architecture, and evaluation protocol. A fair comparison should therefore evaluate alternative modelling strategies on the same held-out test set.

Transfer learning offers an alternative to training an entire network from random initialization. A pretrained model can reuse visual representations learned from a large image collection. MobileNetV2 uses inverted residual blocks and linear bottlenecks to provide an efficient convolutional architecture [3]. A frozen feature extractor allows only a task-specific classification head to learn, whereas fine-tuning permits selected pretrained layers to adapt to the target dataset.

This study compares a baseline CNN, frozen MobileNetV2, and fine-tuned MobileNetV2 on PneumoniaMNIST, a benchmark included in the MedMNIST collection [4]. In addition to threshold-0.5 performance, the revised analysis includes bootstrap confidence intervals, paired differences between models, validation-selected operating thresholds, and a temperature-scaling diagnostic.

The contributions of this work are:

- A three-way comparison of a CNN trained from scratch, frozen MobileNetV2, and fine-tuned MobileNetV2.
- Evaluation on the same held-out test split using classification metrics, ROC-AUC, and confusion-matrix counts.
- Bootstrap confidence intervals and paired comparisons for key test metrics.
- An analysis of validation-selected thresholds and a preliminary calibration assessment, with limitations stated explicitly.

This is an educational benchmark comparison. The models are not intended for clinical diagnosis.

II. Related Work

Kermany *et al.* investigated deep learning for medical diagnosis using retinal and chest X-ray images, including pediatric pneumonia classification [2]. Their work helped demonstrate that convolutional models can learn discriminative patterns from medical images, while also highlighting the importance of carefully constructed training and evaluation procedures.

MobileNetV2 was introduced by Sandler *et al.* as a lightweight convolutional network based on inverted residual blocks and linear bottlenecks [3]. Its architecture is attractive

TABLE I
PNEUMONIAMNIST DATASET SPLIT USED IN THE EXPERIMENT

Split	Normal	Pneumonia	Total
Training	1,214	3,494	4,708
Validation	135	389	524
Test	234	390	624

when computational efficiency is important. In transfer learning, the pretrained convolutional base can initially be frozen and then, where appropriate, selected layers can be fine-tuned on the target task.

The MedMNIST v2 benchmark collection provides standardized, lightweight biomedical image datasets and evaluation settings [4]. PneumoniaMNIST provides a binary pediatric chest X-ray classification task. The present study focuses on a controlled educational comparison between a compact CNN trained from scratch and two MobileNetV2 transfer-learning configurations.

III. DATASET AND PREPROCESSING

A. Dataset

The experiment uses PneumoniaMNIST from MedMNIST, based on pediatric chest X-rays from Guangzhou, China [4], [5]. Each image is labelled Normal (class 0) or Pneumonia (class 1). The experiment loads the dataset with `size=224`, producing inputs configured to 224×224 pixels. The supplied training, validation, and test partitions are retained rather than randomly re-splitting the data.

The training split is imbalanced, with substantially more pneumonia examples than normal examples. Validation data are used for model selection and threshold analysis, while the test split is reserved for final evaluation. The images are pediatric chest X-rays based on the source described in the MedMNIST/PneumoniaMNIST documentation.

B. Preprocessing and Augmentation

Images are loaded as grayscale arrays and converted to floating-point tensors. The baseline CNN rescales pixel intensities from the $[0, 255]$ range to $[0, 1]$. For MobileNetV2, the grayscale channel is repeated three times to match the network input and pixel values are scaled to approximately $[-1, 1]$.

Training-time augmentation includes random rotation, zoom, and contrast adjustment. The configured factors are a rotation factor of 0.05, zoom factor of 0.1, and contrast factor of 0.1. Augmentation is active during training only; validation and test images are not augmented. Class weights calculated from the training labels are approximately 1.939 for Normal and 0.674 for Pneumonia.

IV. METHODOLOGY

A. Baseline CNN

The baseline CNN is trained from scratch. Its input is a single-channel 224×224 image. Four convolutional blocks use 32, 64, 128, and 128 filters, respectively. Each convolution uses a 3×3 kernel, same padding, and ReLU activation, followed by max pooling. Global average pooling is followed by a 128-unit dense layer, dropout with rate 0.5, and a sigmoid output for binary classification. The model has 256,897 trainable parameters.

B. MobileNetV2 Models

Both MobileNetV2 configurations use ImageNet-pretrained weights and replace the original classification head with a task-specific binary head, following the documented application interface and transfer-learning workflow [6], [7]. Images are converted to three channels and processed using MobileNetV2 preprocessing. The frozen configuration keeps the pretrained convolutional base fixed and trains the newly added classification head. Its recorded architecture has 2,259,265 total parameters, of which 1,281 are trainable and 2,257,984 are frozen.

The fine-tuned configuration allows selected pretrained feature layers to adapt to the pediatric chest X-ray task in an additional training stage. It is evaluated separately from the frozen configuration so that the effect of updating pretrained features can be compared on the same test images.

C. Training Procedure

The recorded experiments use TensorFlow/Keras in Google Colab. The baseline CNN and frozen MobileNetV2 configurations use Adam optimization and binary cross-entropy loss. The batch size is 32 and the random seed is 42. Training histories contain 6 epochs for the CNN, 10 for frozen MobileNetV2, and 15 for fine-tuned MobileNetV2. Recorded wall-clock times were approximately 59.0 s, 98.3 s, and 164.8 s, respectively. These times depend on the runtime environment and are not hardware-independent benchmarks.

D. Evaluation Metrics and Thresholding

Pneumonia is treated as the positive class. At the default threshold of 0.5, a predicted score greater than or equal to the threshold is labelled Pneumonia. The confusion matrix records true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP). The metrics are defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Specificity is calculated as $TN/(TN + FP)$. ROC-AUC measures the area under the receiver operating characteristic curve across thresholds.

A separate operating-threshold analysis selected one threshold per model on validation data to target a validation specificity of approximately 0.963. These thresholds were then evaluated on the held-out test set. The threshold analysis is reported separately from the default-threshold results so that the two operating points are not conflated.

V. Results and Discussion

A. Held-Out Test Results at Threshold 0.5

Table II summarizes the recorded test metrics. Fine-tuned MobileNetV2 achieved the highest accuracy, precision, F1-score, specificity, and ROC-AUC of the three models. Frozen MobileNetV2 had the highest recall, 0.992, compared with 0.990 for the fine-tuned model. The small recall difference should be considered alongside the improvement in specificity and the reduction in false positives obtained by fine-tuning.

Relative to the baseline CNN, the frozen model improved accuracy by 0.045 and ROC-AUC by 0.077. Fine-tuning then improved accuracy by a further 0.037 and ROC-AUC by 0.013 relative to the frozen model. Fine-tuning also reduced false positives from 92 to 68, while false negatives increased from 3 to 4. Thus, fine-tuning improved the balance between the two classes in this run, although it did not maximize recall.

B. Bootstrap Confidence Intervals

Bootstrap confidence intervals were generated for the test metrics. Table III reports the available 95% intervals for accuracy, recall, F1-score, and ROC-AUC. The fine-tuned model has the highest point estimates for accuracy, F1-score, and ROC-AUC. These intervals describe uncertainty under the resampling procedure for this test set; they do not replace independent external validation.

Paired comparisons of model predictions produced positive confidence intervals for accuracy, F1-score, and ROC-AUC when fine-tuned MobileNetV2 was compared with the frozen model. The paired recall difference was -0.003 with a reported interval of $[-0.008, 0.000]$; at the reported precision, this interval reaches zero and does not establish a clear recall improvement. Comparisons of frozen MobileNetV2 against the baseline CNN showed positive intervals for all four metrics summarized in the paired analysis.

C. ROC Analysis

Fig. 2 presents the receiver operating characteristic curves for the three models. Fine-tuned MobileNetV2 achieved the highest test ROC-AUC (0.983), followed by frozen MobileNetV2 (0.970) and the baseline CNN (0.893). This indicates better ranking discrimination on this test set across operating thresholds, but it does not establish clinical utility or generalization to other hospitals or populations.

D. Validation Behaviour

The validation summary in Table V shows that fine-tuned MobileNetV2 achieved validation accuracy of 0.979 and ROC-AUC of 0.997. Its validation sensitivity and specificity were 0.992 and 0.941, respectively. Validation results are useful for model selection, but performance on the held-out test set is lower, particularly for the baseline CNN. The individual training histories in Figs. 3–5 show model-specific optimization behaviour. A cross-model comparison of validation curves appears in Fig. 6(a).

E. Validation-Selected Threshold Analysis

A threshold selected on validation data can change the trade-off between sensitivity and specificity. The selected thresholds were 0.777 for the baseline CNN, 0.694 for frozen MobileNetV2, and 0.592 for fine-tuned MobileNetV2. Table VI compares the default threshold of 0.5 with these validation-selected thresholds on the test set.

For fine-tuned MobileNetV2, the validation-selected threshold increased test accuracy from 0.885 to 0.888 and specificity from 0.709 to 0.722, while sensitivity decreased slightly from 0.990 to 0.987. False positives decreased from 68 to 65, while false negatives increased from 4 to 5. Frozen MobileNetV2 showed a larger accuracy and specificity increase, with a small reduction in sensitivity. Thresholds should be chosen on validation data only; the test set is used here to report the resulting operating-point performance.

F. Calibration Analysis

Temperature scaling was evaluated as a post-processing calibration diagnostic, following the general principles of probability calibration and reliability assessment [8], [9]. Table VII reports the temperature values and the available log-loss, Brier-score, and expected calibration error (ECE) values before and after scaling. Lower values are generally preferred for each metric. For fine-tuned MobileNetV2, log-loss decreased from 0.3702 to 0.3421 and the Brier score decreased from 0.0908 to 0.0887, while ECE increased slightly from 0.1048 to 0.1054. For the CNN and frozen MobileNetV2, all three reported metrics worsened after scaling. Thus, the available results do not show a consistent calibration improvement across models.

The exported calibration summary does not state which data split was used to estimate each temperature parameter. To preserve an unbiased test assessment, the temperature must be fitted on validation data only and then applied unchanged to test predictions. The fine-tuned model's reliability diagrams are shown in Fig. 6(b). Until the calibration split is confirmed in the final notebook, the post-scaling figures should be interpreted as exploratory rather than definitive test-set calibration evidence.

VI. Limitations and Responsible Use

Several limitations should be considered. First, the experiment uses one benchmark dataset and one recorded training run per model; results may vary across random seeds and dataset samples. Second, the test set is relatively small and class-imbalanced. Third, the experiment does not include external validation, prospective testing, subgroup analysis, or a clinical reader study. The differences between validation and test performance also warrant further investigation, including verification of patient-level independence and possible image-level dependencies across the supplied splits. Fourth, false-positive predictions remain substantial: the fine-tuned model labels 68 of 234 Normal test images as Pneumonia at threshold 0.5. Fifth, calibration results are provisional until the split used to fit the temperature parameters is verified. Finally, resized benchmark images and an educational training pipeline do not

TABLE II
HELD-OUT TEST RESULTS AT A CLASSIFICATION THRESHOLD OF 0.5

Model	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	TN	FP	FN	TP	Time (min)
CNN (scratch)	0.803	0.785	0.944	0.568	0.857	0.893	133	101	22	368	0.98
MobileNetV2 frozen	0.848	0.808	0.992	0.607	0.891	0.970	142	92	3	387	1.64
MobileNetV2 fine-tuned	0.885	0.850	0.990	0.709	0.915	0.983	166	68	4	386	2.75

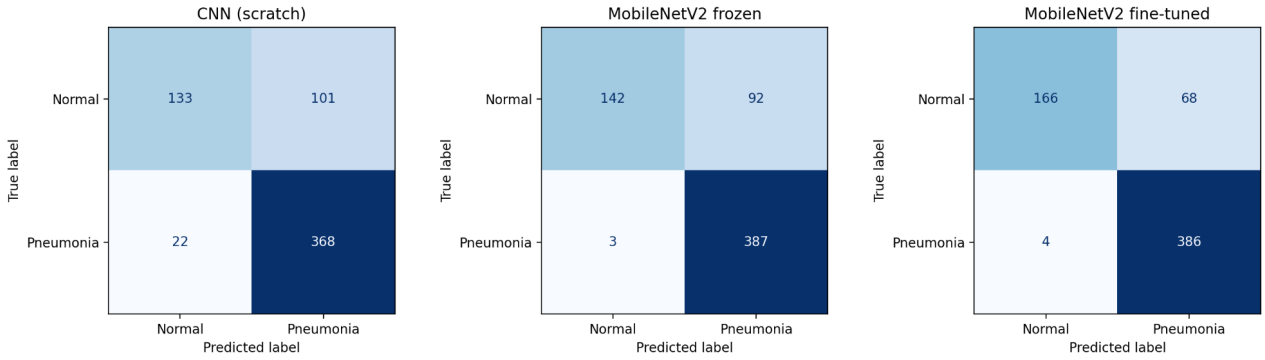


Fig. 1. Confusion matrices on the held-out test set. Rows correspond to true labels and columns to predicted labels. Fine-tuned MobileNetV2 produces fewer false positives than the frozen model, while both retain high pneumonia sensitivity.

TABLE III
TEST METRICS WITH 95% BOOTSTRAP CONFIDENCE INTERVALS

Model	Accuracy [95% CI]	Recall [95% CI]	F1 [95% CI]	ROC-AUC [95% CI]
CNN (scratch)	0.803 [0.772, 0.832]	0.944 [0.920, 0.966]	0.857 [0.833, 0.880]	0.893 [0.869, 0.916]
MobileNetV2 frozen	0.848 [0.819, 0.877]	0.992 [0.982, 1.000]	0.891 [0.869, 0.912]	0.970 [0.957, 0.981]
MobileNetV2 fine-tuned	0.885 [0.859, 0.910]	0.990 [0.978, 0.998]	0.915 [0.895, 0.934]	0.983 [0.974, 0.990]

TABLE IV
PAIRED DIFFERENCES IN TEST METRICS (DIFFERENCE AND 95% CONFIDENCE INTERVAL)

Metric	Frozen MobileNetV2 − CNN	Fine-tuned − Frozen MobileNetV2
Accuracy	+0.045 [+0.016, +0.074]	+0.037 [+0.019, +0.056]
Recall	+0.049 [+0.028, +0.070]	−0.003 [−0.008, +0.000]
F1-score	+0.034 [+0.015, +0.054]	+0.024 [+0.012, +0.037]
ROC-AUC	+0.077 [+0.057, +0.098]	+0.013 [+0.007, +0.019]

TABLE V
VALIDATION-SET SUMMARY AT THE DEFAULT THRESHOLD

Model	Acc.	AUC	Sens.	Spec.	FN	FP
CNN (scratch)	0.884	0.941	0.903	0.830	37	23
MobileNetV2 frozen	0.967	0.994	0.987	0.911	5	12
MobileNetV2 fine-tuned	0.979	0.997	0.992	0.941	3	8

represent the full complexity of clinical radiology workflows. Transparent reporting of data sources, model development, and internal versus external testing is also emphasized by medical-imaging AI reporting guidance [10]. The models should not be used to make medical decisions.

VII. CONCLUSION AND FUTURE WORK

This study compared a baseline CNN trained from scratch with frozen and fine-tuned MobileNetV2 models for binary classification of pediatric chest X-ray images. At a threshold of 0.5, fine-tuned MobileNetV2 achieved the strongest overall test performance, with accuracy 0.885, F1-score 0.915, ROC-AUC 0.983, and specificity 0.709. It reduced false positives relative to the frozen model from 92 to 68, while maintaining recall of 0.990. A validation-selected threshold of 0.592 modestly increased its test accuracy to 0.888 and specificity to 0.722, with sensitivity of 0.987. The results indicate that fine-tuning improved the balance of errors in this recorded experiment, but high recall alone is not sufficient evidence of reliability.

Future work should include repeated runs with multiple random seeds, fully documented bootstrap procedures, ex-

TABLE VI
TEST PERFORMANCE AT THE DEFAULT AND VALIDATION-SELECTED THRESHOLDS

Model	Val. thr.	Acc. @0.5	Acc. @thr.	Sens. @0.5	Sens. @thr.	Spec. @0.5	Spec. @thr.	FN @thr.	FP @thr.
CNN (scratch)	0.777	0.803	0.819	0.944	0.864	0.568	0.744	53	60
MobileNetV2 frozen	0.694	0.848	0.883	0.992	0.982	0.607	0.718	7	66
MobileNetV2 fine-tuned	0.592	0.885	0.888	0.990	0.987	0.709	0.722	5	65

TABLE VII
CALIBRATION METRICS BEFORE AND AFTER TEMPERATURE SCALING

Model	T	LogLoss before	LogLoss after	Brier before	Brier after	ECE before	ECE after
CNN (scratch)	0.84	0.4635	0.4934	0.1472	0.1510	0.1151	0.1259
MobileNetV2 frozen	0.71	0.3834	0.4744	0.1123	0.1201	0.1413	0.1435
MobileNetV2 fine-tuned	1.12	0.3702	0.3421	0.0908	0.0887	0.1048	0.1054

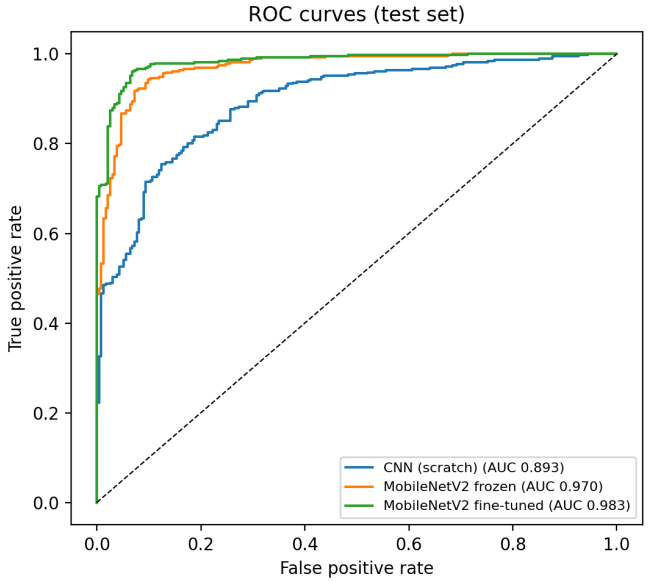


Fig. 2. ROC curves for the three models on the held-out test set.

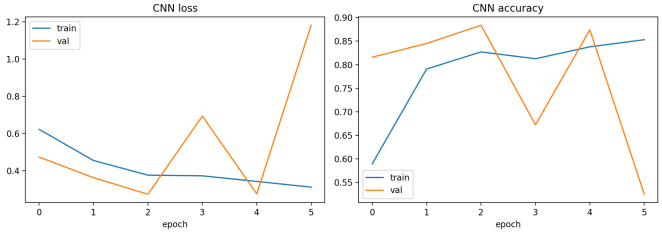


Fig. 3. Training and validation loss and accuracy for the baseline CNN.

ternal validation on independently sourced data, verification of patient-level independence, and more systematic threshold and calibration analysis. Calibration parameters should be learned on validation data and evaluated unchanged on the test set. Explainability methods such as Grad-CAM may also be examined with domain experts [11]. Any future clinical application would require appropriate medical, ethical, and regulatory review.

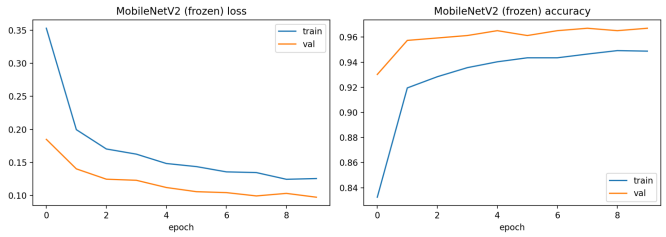


Fig. 4. Training and validation loss and accuracy for frozen MobileNetV2.

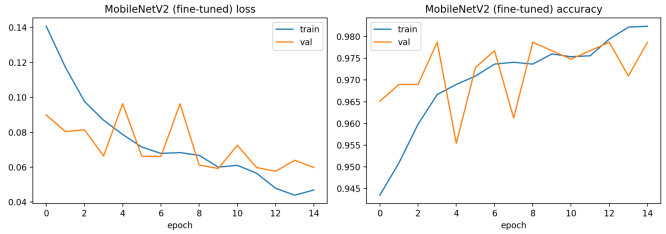
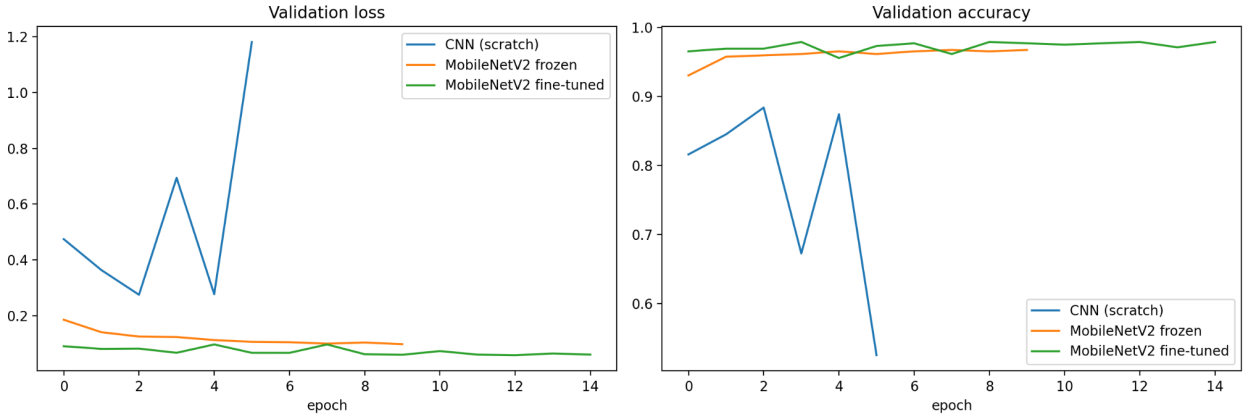
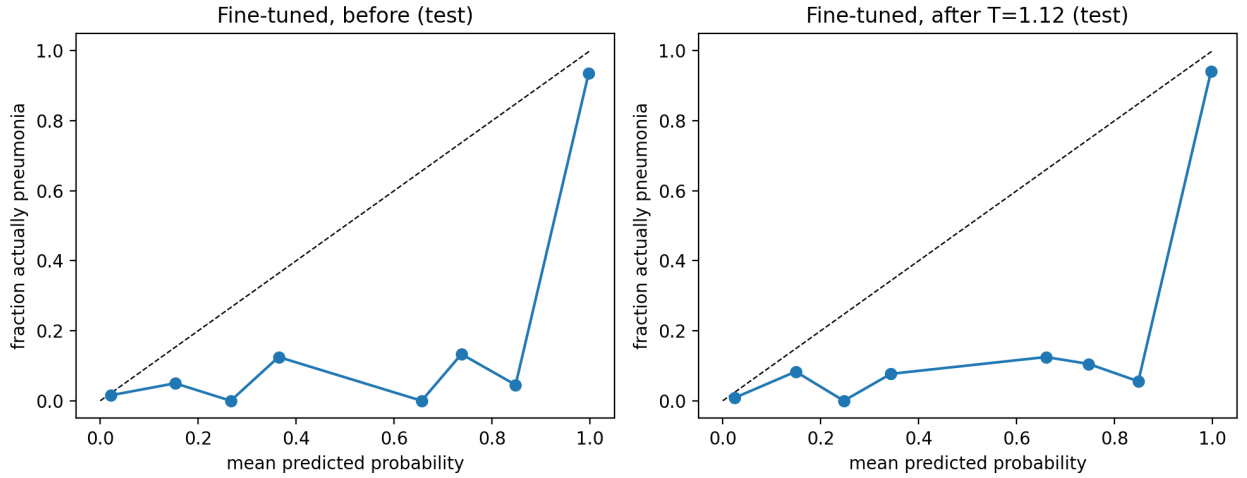


Fig. 5. Training and validation loss and accuracy for fine-tuned MobileNetV2.



(a) Validation loss and accuracy histories for the three models.



(b) Fine-tuned MobileNetV2 reliability diagrams before and after temperature scaling with $T = 1.12$.

Fig. 6. Final validation and calibration diagnostics. The reliability plot is a binned diagnostic and does not by itself establish clinical probability calibration.

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